

Revising All we have learnt.

Population

$$\mu = \frac{N}{N} \sum_{i=1}^N (x_i)$$

$$\sigma^2 = \frac{N}{N} \sum_{i=1}^N (x_i - \mu)^2$$

How far a
Data point is away
from the mean.
 $\sqrt{\sigma^2} \rightarrow$ Population
Standard deviation

$$\sigma = \sqrt{\text{Population Variance}}$$

Sample

n

$$\bar{x} = \frac{n}{n} \sum_{i=1}^n \frac{(x_i)}{n}$$

$$S^2 = \frac{n}{n-1} \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}$$

$S = \sqrt{\text{Sample Variance}}$
 $\downarrow$
Sample Standard Deviation